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IKB42603 • Cloud Security

WEEK 3 • LESSON 3

Secure Isolation of Physical & Logical Infrastructure

Multi-tenancy · Compute · Network · Storage isolation



What you will be able to do



1. Understand isolation in the cloud

Why separation is critical on shared infrastructure.

2. Explain the three isolation types

Compute, network and storage isolation.

3. Identify common attack vectors

VM escape, eavesdropping, data remanence and more.

4. Apply secure isolation strategies

Hypervisor hardening, segmentation, encryption.

5. Understand multi-tenancy risks

And the controls that keep tenants safely separated.

Curriculum alignment — CLO · PLO · MQF · clusters

CLO 1

Examine how cloud computing security concepts are applied to solve real-world technical problems.

CLO 2

Construct secure cloud computing operations that safeguard data integrity.



Programme outcome

PLO1 · PLO2
MQF i · iii(a)
Bloom C3 · P3



Values & skills

SC1 · System Thinking
SC8 · Problem-Solving
VBE3 · Integrity



Sustainability (SDG)

SDG 9.1
Resilient, inclusive infrastructure



Assessment

Test
Lab Assignment

Constructive alignment: CLO → PLO → MQF, with VBE (values), SC (skills) and SDG clusters, assessed as shown.

01

PART 1

Why Isolation Matters: Multi-Tenancy



DEFINITION

What is secure isolation?

The mechanisms that keep tenants separated on shared infrastructure — so no customer can access, disturb or observe another's workloads.

- Separation between tenants (customers)
- Prevents unauthorised access & data leakage
- Shared but securely separated

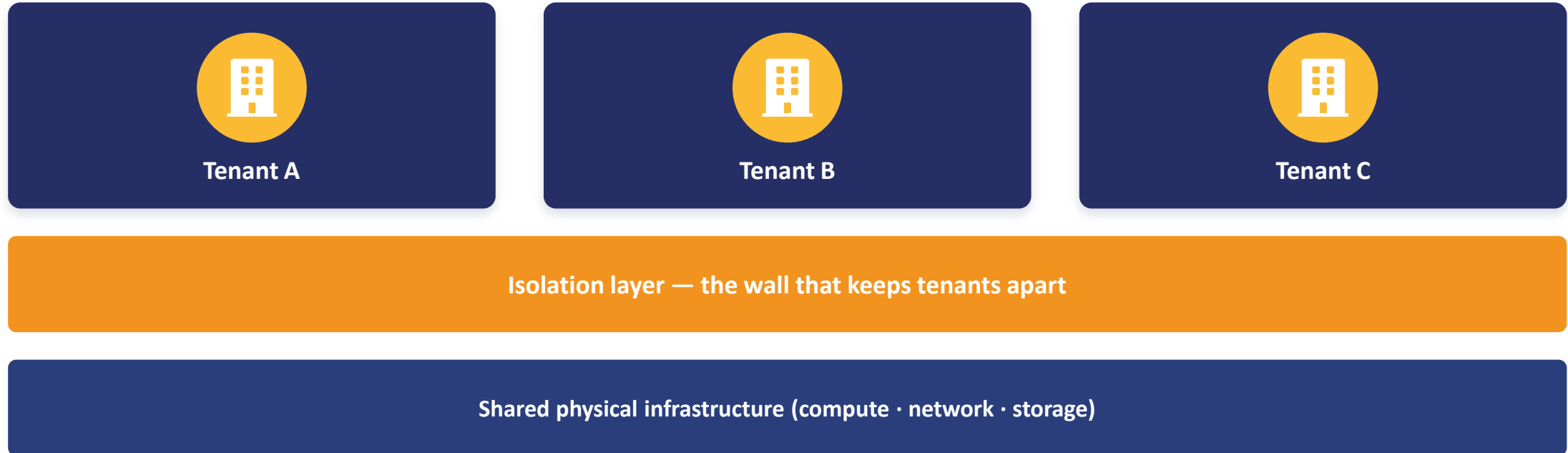
Enforced across three dimensions: *compute · network · storage.*



Shared, but securely separated.

Resource pooling makes the cloud cheap — and creates the isolation challenge. A failure here breaks the trust the whole model depends on.

Multi-tenancy — many customers, one infrastructure



Like an apartment block: *many families, one building — but locked doors and solid walls mean neighbours never see each other.*

Three dimensions of isolation



Compute isolation

Separate processing — CPU, memory, workloads.

VMs · containers · hypervisor



Network isolation

Separate traffic between tenants.

VLANs · VPCs · firewalls



Storage isolation

Separate stored data between tenants.

Logical separation · encryption · access control

An attacker needs only ONE dimension to be weak — defenders must secure all three.

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PART 2

Compute Isolation



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Compute isolation — separating workloads



Virtual machines (VMs)

Each tenant gets a full virtual computer with its own OS.

- Isolated by the hypervisor
- Strong, hardware-level boundary
- Higher overhead, slower to start



Containers

Package an app with its dependencies; share the host kernel.

- Lightweight, fast, high density
- Thinner isolation (shared kernel)
- Great for microservices

Hypervisor security — harden the foundation



The hypervisor is the isolation boundary — compromise it and every VM on the host falls.

Keep it minimal: *a lean hypervisor with few services has a smaller attack surface.*



Access restrictions

Tightly limit reach to the management interface.



Monitoring

Watch closely — an attack here is catastrophic.



Updates & patching

Hypervisor flaws are high-value targets.

Container security & compute risks



Container security

- Lightweight — faster than VMs
- Shares the host OS kernel
- Weaker isolation than VMs
- Risk: shared kernel + misconfiguration

Compute isolation risks



VM escape — Break out of a VM to reach the host.



Hypervisor flaws — Vulnerabilities in the critical layer.



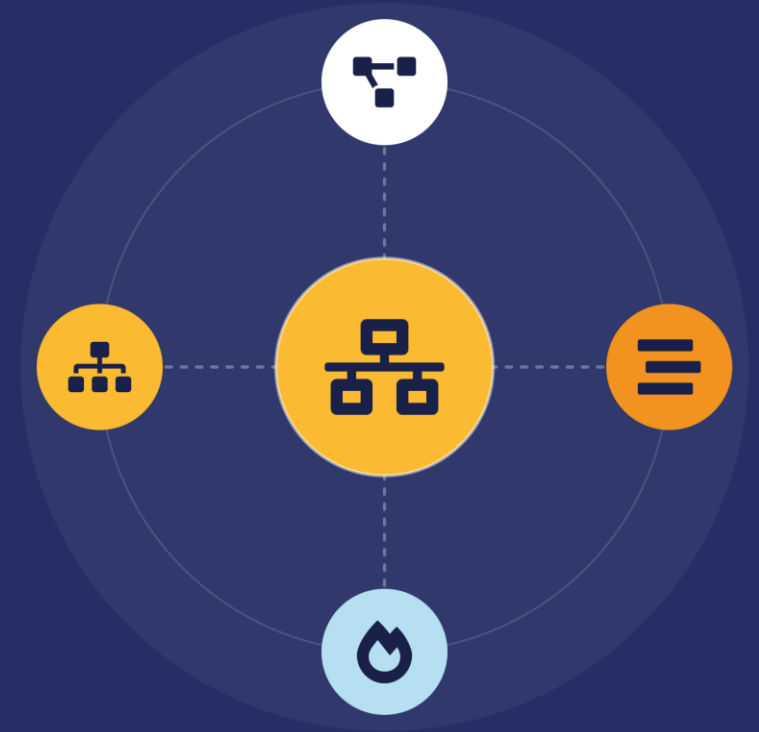
Side-channel attacks — Infer secrets via timing / cache / power.

Defence: *patch relentlessly, least-privilege containers, add a VM boundary for sensitive workloads.*

03

PART 3

Network Isolation



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Network isolation — segmenting traffic

Segment traffic so unauthorised communication between tenants — and within a tenant — is prevented.



VLANs

Logically divide one physical network into separate domains.



Virtual Private Clouds

A private, isolated network slice per tenant.



Firewalls & security groups

Rules controlling what traffic is allowed where.

Principle: *default-deny — permit only the flows that are needed (least privilege for the network).*

Segmentation: benefits and risks



Techniques

- **VLANs** — Classic logical division
- **SDN** — Programmable, policy-driven network
- **Micro-segmentation** — Isolate each individual workload



Risks

- **Packet sniffing** — Captures unencrypted traffic
- **Man-in-the-middle** — Intercepts & alters comms
- **Misconfigured firewalls** — An accidental 'allow all'

Benefit: *segmentation contains lateral movement — it limits how far an attack can spread.*

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PART 4

Storage Isolation



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Storage isolation — separating data



Logical separation

Namespaced, separately-addressed data per tenant — the primary wall.



Encryption

Per-tenant keys — one tenant's key never unlocks another's data.



Access control

Only authorised identities reach each tenant's storage (IAM).

Defense in depth for data: *the wall, the safe inside the wall, and the guard at the door.*

Data remanence & storage risks



Data remanence: 'deleted' data often lingers on freed blocks — and may resurface for the next tenant.

Defence: *secure overwrite, or cryptographic erasure — destroy the key and the data is gone.*



Data leakage

Weak logical separation exposes data.



Unauthorised access

Misconfigured storage permissions.



Improper deletion

Remnants recoverable by others.

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PART 5

Strategy, Case Study & Practice



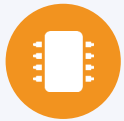
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Secure isolation strategy & threats

Isolation strategy — four controls together



Multi-tenancy controls



Virtualization security



Network segmentation



Storage protection

The combined threat picture



VM escape (*compute*)



API exploitation (*compute*)



Network eavesdropping (*network*)



Insider threats (*all layers*)



Misconfiguration (*all layers*)

Case study — the cross-tenant leak



A cloud provider misconfigures isolation. As a result, one tenant is able to reach another tenant's data — a cross-tenant leak, the worst-case isolation failure.

What failed?

A gap in one dimension — storage separation, a firewall rule, a hypervisor/container weakness, or remanence

How to prevent?

Strong logical separation · per-tenant encryption · hardened virtualization · segmentation · monitoring

Best practices — enforcing isolation



Use strong isolation

Across compute, network and storage.



Patch relentlessly

Hypervisors and container hosts first.



Monitor infrastructure

Detect isolation failures early.



Apply least privilege

On the network and on storage.



Encrypt sensitive data

Per-tenant keys protect crossed boundaries.



Securely delete data

Defeat data remanence.

Key takeaways



1 Isolation is the foundational pillar — keep tenants separated; its failure breaks the whole trust model.



2 It must hold across three dimensions — compute, network, storage. One weak point is enough for an attacker.



3 Multi-tenancy creates the risk; controls are hardened virtualization, segmentation, and storage encryption.



4 Signature threats: VM escape, eavesdropping, data remanence — with misconfiguration the recurring root cause.



Discussion & Q&A

- Which isolation layer — compute, network or storage — do you think is most vulnerable, and why?
Are containers less secure than virtual machines? Justify your answer.

Next week — Lesson 4: Data Protection for Cloud Infrastructure & Services (encryption · PKI · key management)